Roll No. : .....

ODD END SEMESTER EXAMINATION 2024

Name of the Program: M. Tech (CSE) **Semester: I**

Course Name: Research Methodology and IPR **Course Code: MCS-114**

Time: 3 Hours **Maximum Marks: 100**

Note:

i. All questions are compulsory.
ii. Answer any two questions among a, b, and c in each main question.
iii. The total marks for each main question are twenty.

Q1	**(20 Marks)**	
(a)	Examine the criteria and characteristics that make a good research problem. What qualities should researchers look for when selecting a research problem, and how do these characteristics influence the overall quality and success of the study?	CO1
(b)	How do the scope and objectives of a research problem shape the direction of a study? Explore the relationship between a research problem's scope, its objectives, and the feasibility of carrying out meaningful research.	CO2
(c)	What is the meaning of a research problem, and why is it crucial to clearly define it before embarking on any research? Discuss the various sources from which research problems can arise and provide examples.	CO2
Q2	**(20 Marks)**	
(a)	Ethics is important in research, especially when working with people or sensitive information. What are the key ethical rules researchers must follow? How can they make sure their work is ethical, and what could happen if they don't follow these rules?	CO3
(b)	Why is literature review important in research? How does it help find research problems and gaps in knowledge? What steps should researchers follow to do a good literature review?	CO2
(c)	What are the ethical issues involved in using AI to detect plagiarism in academic writing? How can AI help maintain integrity, and what problems like false positives and privacy concerns should be addressed?	CO3
Q3	**(20 Marks)**	
(a)	Describe the process of developing a research proposal. What are the critical elements to include in a research proposal, and how do you ensure that your proposal is both well-supported and feasible for the intended audience?	CO1
(b)	Discuss the challenges and best practices associated with writing technical reports in a collaborative team setting. How can teams ensure a cohesive and well-written report?	CO4
(c)	Explore the significance of an academic or research presentation. What strategies should researchers use to effectively present their findings to a review committee, and how is a research proposal typically assessed during a review?	CO2
Q4	**(20 Marks)**	
(a)	Explain the concept of Intellectual Property Rights (IPR) and describe its various types, including the duration of protection for each.	CO5
(b)	What are the key challenges in detecting and prosecuting IPR infringements in the digital age, particularly with regard to online platforms and digital media?	CO5
(c)	Discuss the Patent Cooperation Treaty (PCT) and its significance in facilitating international patent protection. What are the advantages of using the PCT system for inventors seeking patent protection in multiple countries, and how does the process work?	CO6
Q5	**(20 Marks)**	

(a)	Explain how Geographical Indications (GIs) help protect products that are unique because of where they come from. How do GIs affect local economies, culture, and trade? Give examples of famous GIs and why they are important.	CO5
(b)	What are the recent changes in Intellectual Property Rights (IPR) for biology and software? How are IPR rules adapting to new challenges in these areas? What are the ethical, legal, and practical effects of these changes?	CO6
(c)	What are patent rights in intellectual property law, and how do they vary in different countries? What factors affect how patents are enforced and protected? Give examples of cases where patent rights were changed or challenged because of new technologies.	CO6